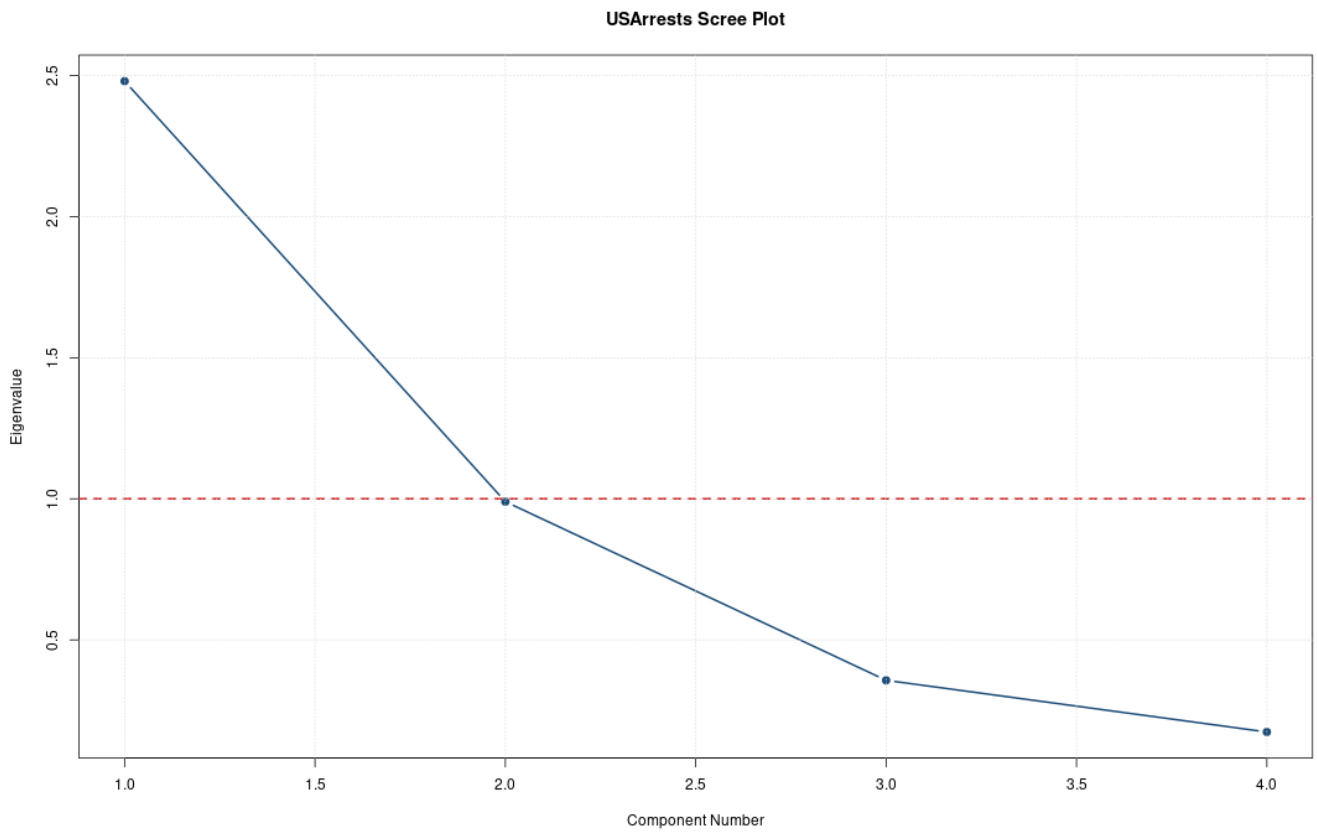
Practice Questions Report: Factor Analysis and Hierarchical Clustering

This report solves the four practice tasks and includes required plots.

1) Factor Analysis on USArrests

Plot



Code Snippet

```
library(psych)

usarrests_scaled <- scale(USArrests)
usarrests_eigs <- eigen(cor(usarrests_scaled))$values

plot(usarrests_eigs, type = "b", pch = 19,
    xlab = "Component Number", ylab = "Eigenvalue",
    main = "USArrests Scree Plot")
abline(h = 1, col = "red", lty = 2)

fa_us <- fa(usarrests_scaled, nfactors = 2, rotate = "varimax", fm = "ml")
print(fa_us$loadings, cutoff = 0.30)
```

How it works (short)

Factor analysis models observed variables using a smaller number of latent factors. We standardized variables, used a correlation matrix, and extracted 2 factors with varimax rotation. Proportion variance explained (F1, F2): 0.489, 0.214

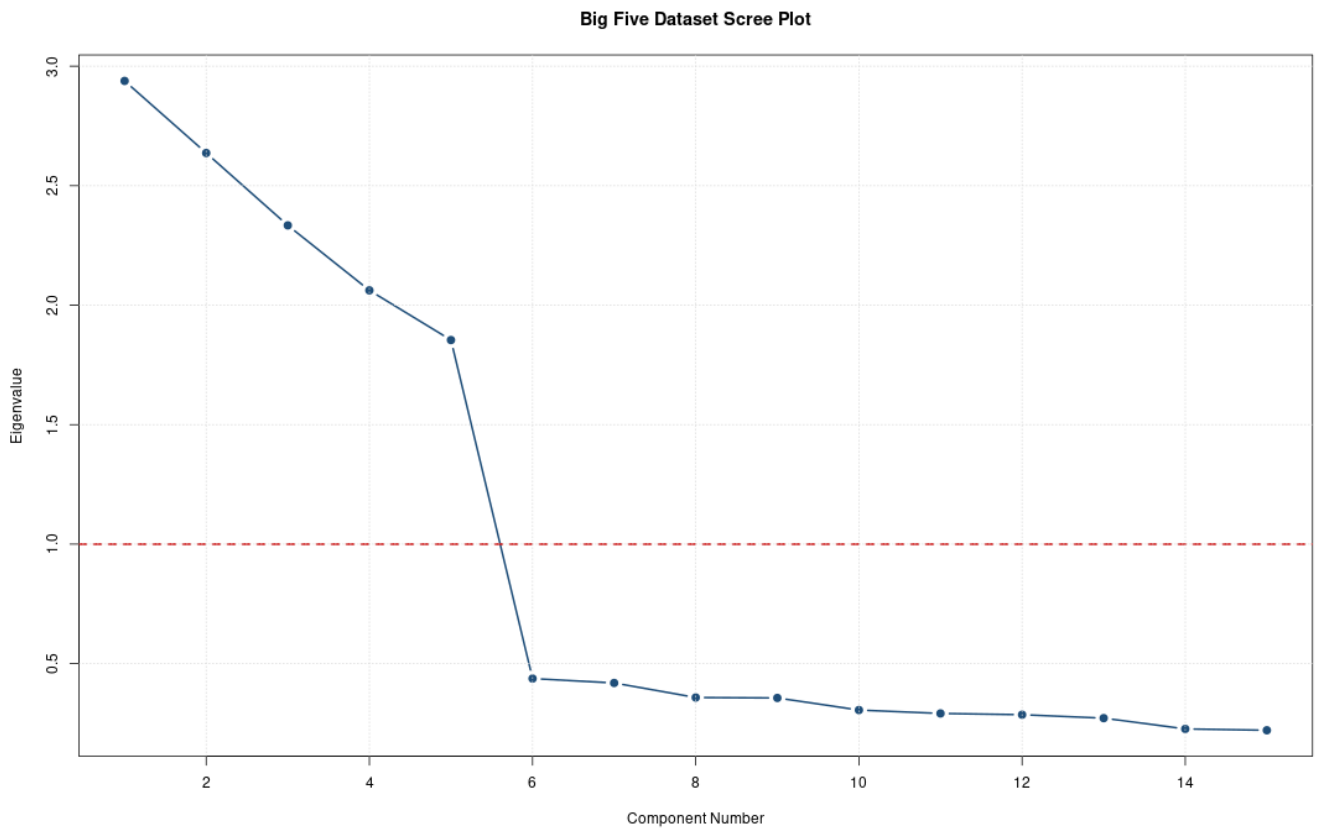
Top loading variables:

- ML1: Murder, Assault
- ML2: UrbanPop, Rape

Saved loadings table: `plots/usarrests_loadings.csv`

2) Factor Analysis on big_five_personality_teaching_dataset

Plot



Code Snippet

```
library(readxl)
library(psych)

bigfive <- read_excel("big_five_personality_teaching_dataset.xlsx")
bigfive_num <- as.data.frame(bigfive)[, sapply(bigfive, is.numeric), drop =
FALSE]
bigfive_num <- na.omit(bigfive_num)
bigfive_scaled <- scale(bigfive_num)
```

```
bigfive_eigs <- eigen(cor(bigfive_scaled))$values
plot(bigfive_eigs, type = "b", pch = 19,
    xlab = "Component Number", ylab = "Eigenvalue",
    main = "Big Five Scree Plot")
abline(h = 1, col = "red", lty = 2)

fa_big <- fa(bigfive_scaled, nfactors = 5, rotate = "varimax", fm = "ml")
print(fa_big$loadings, cutoff = 0.30)
```

How it works (short)

The Big Five items were reduced into 5 latent factors (Extraversion, Agreeableness, Conscientiousness, Neuroticism, Openness style structure) using ML factor analysis with varimax rotation. Proportion variance explained (F1..F5): 0.141, 0.138, 0.138, 0.134, 0.132

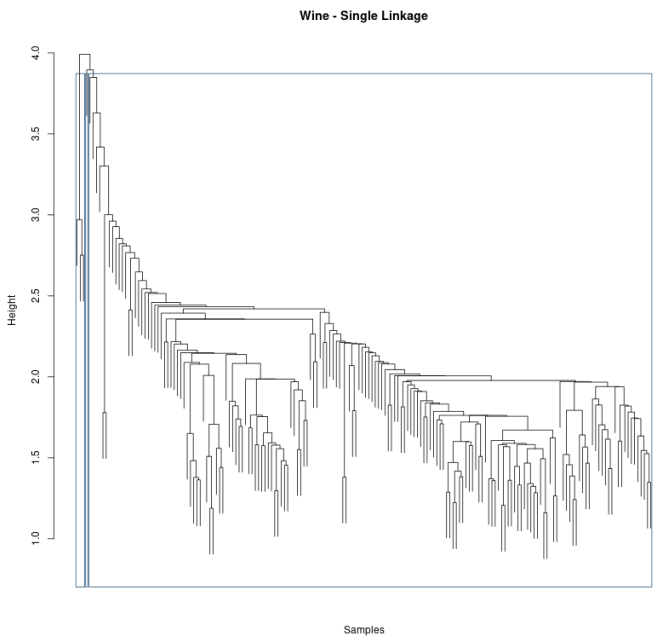
Top loading variables:

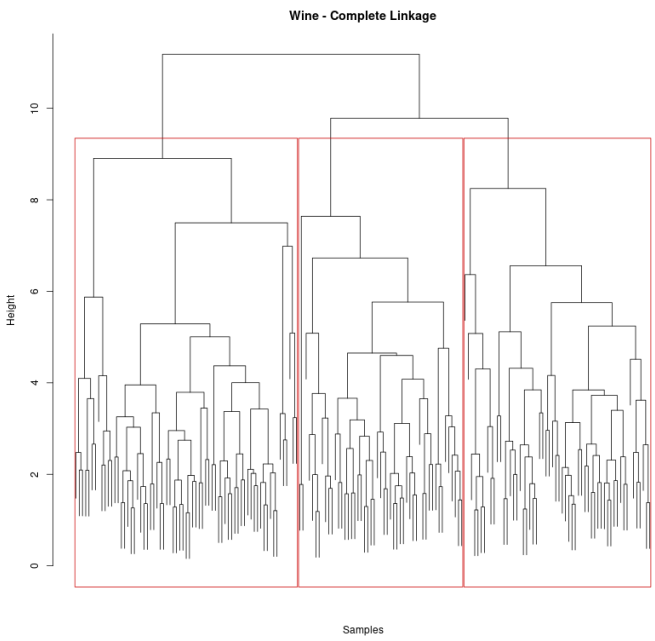
- ML1: A2_Cooperative, A3_Forgiving
- ML2: C1_Organized, C2_Disciplined
- ML3: N2_Worry, N3_Moody
- ML4: O2_Imaginative, O1_Creative
- ML5: E3_Energetic, E2_Sociable

Saved loadings table: `plots/bigfive_loadings.csv`

3) Hierarchical Clustering on Wine Dataset (Single vs Complete)

Plot





Code Snippet

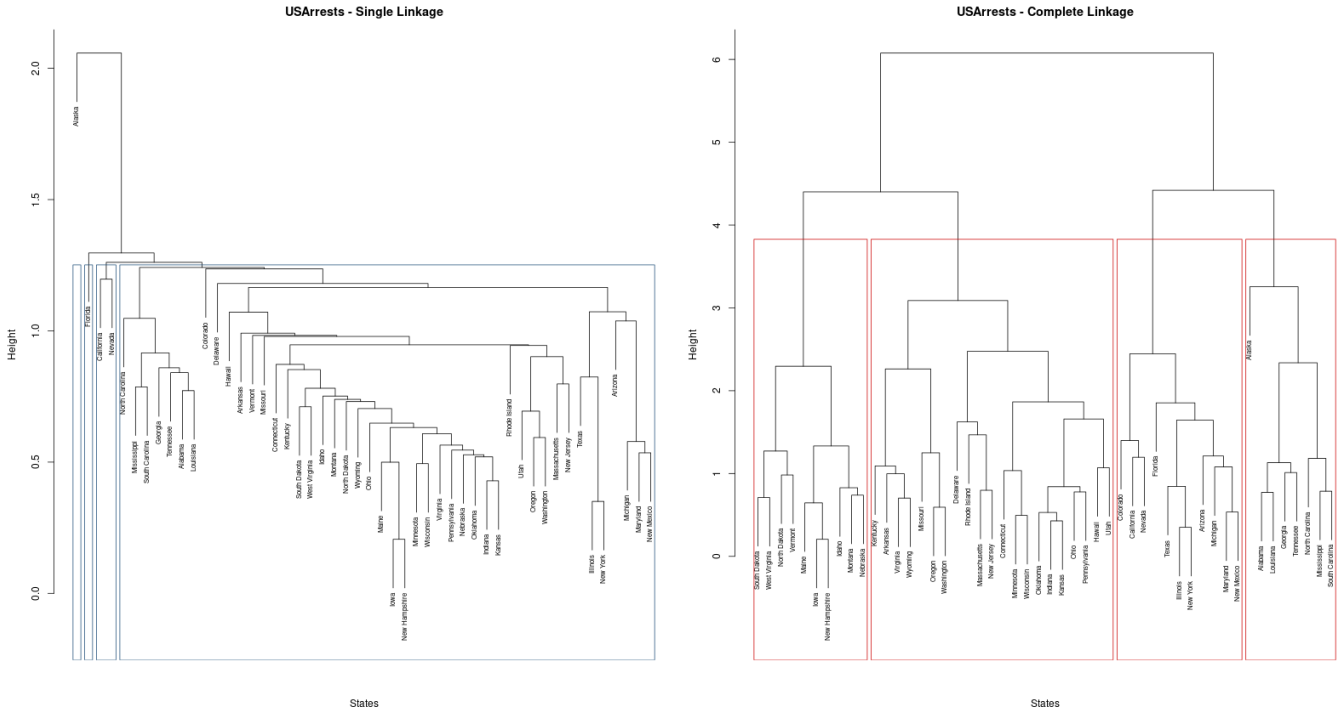
```
# wine_features: numeric wine measurements (class column removed)
wine_scaled <- scale(wine_features)
wine_dist <- dist(wine_scaled)

wine_hc_single <- hclust(wine_dist, method = "single")
wine_hc_complete <- hclust(wine_dist, method = "complete")

par(mfrow = c(1, 2))
plot(wine_hc_single, labels = FALSE, main = "Wine - Single Linkage")
rect.hclust(wine_hc_single, k = 3, border = "blue")
plot(wine_hc_complete, labels = FALSE, main = "Wine - Complete Linkage")
rect.hclust(wine_hc_complete, k = 3, border = "red")
```

How it works (short)

Hierarchical clustering starts with each sample as its own cluster and merges clusters step-by-step using distance rules. Single linkage uses nearest points (can chain clusters), while complete linkage uses farthest points (more compact clusters).

4) Cluster US States by Crime Rates (USArrests)

Plot



Code Snippet

```
usarrests_scaled <- scale(USArrests)
us_dist <- dist(usarrests_scaled)

us_hc_single <- hclust(us_dist, method = "single")
```

```
us_hc_complete <- hclust(us_dist, method = "complete")

par(mfrow = c(1, 2))
plot(us_hc_single, cex = 0.65, main = "USArrests - Single Linkage")
rect.hclust(us_hc_single, k = 4, border = "blue")
plot(us_hc_complete, cex = 0.65, main = "USArrests - Complete Linkage")
rect.hclust(us_hc_complete, k = 4, border = "red")

us_clusters <- data.frame(
    State = rownames(USArrests),
    Single_Linkage_Cluster = cutree(us_hc_single, k = 4),
    Complete_Linkage_Cluster = cutree(us_hc_complete, k = 4)
)
head(us_clusters)
```

How it works (short)

After standardizing crime variables, hierarchical clustering groups states with similar crime profiles. Single linkage may connect clusters early; complete linkage usually gives tighter and more separated groups.

Saved cluster membership table: `plots/usarrests_clusters.csv`

To rerun everything: `Rscript solve_practice_questions.R`